■ Developer Mock Test Results

Test ID:	acca68ab-512c-44...	**Student ID:**	N/A
Test Type:	Developer Assessment	**Date:**	2026-03-04 16:39:23
Overall Score:	11/25 (44.0%)	**Performance:**	■ Average
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	6/10	60.0%	■ Pass
■ MCQ/Theory	5/10	50.0%	■ Pass
■ Coding	0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (6/10 - 60.0%)

■ Q1. The average of 4 numbers is 15. If one of the numbers is 20, what is the sum of the remaining 3 numbers?

Your Answer: 45
■ *Correct! Well done.*

■ Q2. A mixture is made of 2 parts water and 3 parts juice. What is the ratio of water to the total mixture?

Your Answer: 1/2
Correct Answer: 2/5
■ *The correct answer is 2/5 because the mixture is made of 2 parts water and 3 parts juice, making a total of 2 + 3 = 5 parts. The ratio of water to the total mixture is then 2/5. The user's likely mistake was assuming the ratio of water to juice (2:3) was the same as the ratio of water to the total mixture, which is not the case, as shown by the calculation 2/(2+3) = 2/5.*

■ Q3. If a number is decreased by 20% and then increased by 20%, what is the overall percentage change?

Your Answer: 4%
Correct Answer: -4%
■ *The correct answer is -4% because when a number is decreased by 20% and then increased by 20%, the overall effect is a decrease. The key calculation step is: if we start with 100, decreasing it by 20% gives 80, and then increasing 80 by 20% gives 96, which is 4% less than the original 100. The user's likely mistake was not considering the effect of the percentage changes on the reduced value, rather than the original value.*

■ Q4. If 8 workers can build a house in 12 days, how many days will it take for 6 workers to build the same house?

Your Answer: 16 days
■ *Correct! Good understanding.*

■ Q5. The average of 5 numbers is 20. If one of the numbers is 30, what is the sum of the remaining 4 numbers?

Your Answer: 60
■ *Correct! Good understanding.*

■ Q6. A shopkeeper buys an item at Rs.50 and sells it at a 15% profit. What is the selling price?

Your Answer: Rs.57.50
■ *Excellent! Right answer.*

■ Q7. A product is on sale for 20% off its original price. If the original price is $100, how much will you pay for the product after the discount?

Your Answer: $80
■ *Correct! Well done.*

■ Q8. The average of 2 numbers is 18. What is the sum of the 2 numbers?

Your Answer: 32
Correct Answer: 36
■ *The correct answer is 36 because the average of 2 numbers is calculated by dividing their sum by 2. The key calculation step is: sum of 2 numbers = average * 2 = 18 * 2 = 36. The user's likely mistake was a calculation error, as they chose 32 instead of the correct result of the calculation, which is 36.*

■ Q9. If it is true that all A are B and some B are not C, what can be concluded about A and C?

Your Answer: Some A are C
■ *Excellent! Right answer.*

■ Q10. A store buys a product for $25 and sells it for $30. What is the profit percentage?

Your Answer: 10%
Correct Answer: 20%
■ *To find the profit percentage, we need to calculate the profit first: Profit = Selling price - Cost price = $30 - $25 = $5. Then, we calculate the profit percentage: (Profit / Cost price) * 100 = ($5 / $25) * 100 = 20%. The user's answer of 10% is likely incorrect because they may have miscalculated the profit percentage or misunderstood the formula.*

■ MCQ SECTION (5/10 - 50.0%)

■ Q1. What is the purpose of the throw keyword in Java?

Your Answer:

```
To catch an exception explicitly
```

Correct Answer:

```
To throw an exception explicitly
```

■ *The correct answer is "To throw an exception explicitly" because the `throw` keyword in Java is used to explicitly throw an exception from a method or a block of code. This allows programmers to manually trigger an exception when a certain condition occurs, giving them more control over error handling. The user's answer, "To catch an exception explicitly", is incorrect because catching an exception is done using the `try-catch` block, not the `throw` keyword.*

■ Q2. What is method overloading in Java?

Your Answer: Multiple methods with the same name but different parameters
■ *Correct! Well done.*

■ Q3. What is the purpose of the Java compiler?

Your Answer: To compile Java code

■ *Excellent! Right answer.*

■ Q4. What is the design of Java programs that makes them easy to maintain?

Your Answer: Structured and class-based

■ *Correct! Good understanding.*

■ Q5. What is the basic syntax of Java?

Your Answer:

```
Variables, data types, control structures, methods, and functions
```

Correct Answer:

```
Variables, data types, control structures, methods, classes, and objects
```

■ *The correct answer is A) Variables, data types, control structures, methods, classes, and objects because it encompasses the fundamental elements that make up the basic syntax of Java. The user's answer, B) Variables, data types, control structures, methods, and functions, is incorrect because Java is an object-oriented language that organizes code into classes and objects, whereas "functions" is a term more commonly used in procedural programming. The user's mistake likely stems from confusing Java's object-oriented nature with procedural programming concepts.*

■ Q6. What are the basic data types in Java?

Your Answer: int, float, char, String

Correct Answer: int, double, char, String

■ *The correct answer is A) int, double, char, String because these are indeed the basic data types in Java, where 'int' is used for integers, 'double' for decimal numbers, 'char' for characters, and 'String' for sequences of characters. The user's answer, C) int, float, char, String, is incorrect because while 'float' is a data type in Java, 'double' is more commonly used for decimal numbers and is considered a basic data type. The user likely mistakenly chose 'float' over 'double', which is the correct basic data type in this context.*

■ Q7. What is concurrency in Java?

Your Answer: The ability of a program to execute multiple threads or processes sequentially

Correct Answer: The ability of a program to execute multiple threads or processes concurrently

■ *The correct answer is A) The ability of a program to execute multiple threads or processes concurrently, because concurrency in Java refers to the ability of a program to execute multiple threads or processes simultaneously, improving responsiveness and system utilization. The user's answer, C) The ability of a program to execute multiple threads or processes sequentially, is incorrect because it describes a sequential execution, which is the opposite of concurrency. The user's likely mistake is misunderstanding the concept of concurrency, which involves simultaneous execution, not sequential execution.*

■ Q8. What is the purpose of type casting in Java?

Your Answer: To reduce code complexity

Correct Answer: To convert one data type to another

■ *The correct answer is "To convert one data type to another" because type casting in Java is used to explicitly convert the data type of a variable from one type to another, allowing for more flexibility in programming. The user's answer, "To reduce code complexity", is incorrect because while type casting may be used to simplify certain code scenarios, its primary purpose is data type conversion. The user likely mistook the potential benefit of type casting for its fundamental purpose.*

■ Q9. What type of programming language is Java?

Your Answer: High-level, object-oriented

■ *Well done!*

■ Q10. What is the purpose of control structures in Java?

Your Answer: To control program flow

■ *Excellent! Right answer.*

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a program that reads a string of words and prints the frequency of each word. **Input Format:** - Line 1: Input string (string) **Output Forma...

Your Answer:

```
import java.util.Scanner;
import java.util.LinkedHashMap;
import java.util.Map;

public class WordFrequency {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine().trim();
        scanner.close();

        // LinkedHashMap preserves insertion order (first appearance of each word)
        Map<String, Integer> freq = new LinkedHashMap<>();

        for (String word : input.split("\\s+")) {
            freq.put(word, freq.getOrDefault(word, 0) + 1);
        }

        StringBuilder result = new StringBuilder();
        for (Map.Entry<String, Integer> entry : freq.entrySet()) {
            if (result.length() > 0) result.append(" ");
            result.append(entry.getKey()).append(":").append(entry.getValue());
        }

        System.out.println(result);
    }
}
```

Correct Answer:

```
import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String[] words = input.split("\\s+");

        Map<String, Integer> wordFrequency = new HashMap<>();

        for (String word : words) {
            wordFrequency.put(word, wordFrequency.getOrDefault(word, 0) + 1);
        }

        for (Map.Entry<String, Integer> entry : wordFrequency.entrySet()) {
            System.out.print(entry.getKey() + ":" + entry.getValue() + " ");
        }

        scanner.close();
    }
}
```

■ *The code failed due to a syntax error, as indicated by the "SyntaxError: invalid syntax" message. The error message specifically points to the line "public class WordFrequency {" which suggests that the Python interpreter is being used to run*

Java code, causing the syntax error. This discrepancy between the code's language (Java) and the interpreter's language (Python) is the root cause of the failure.

■ Q2. Write a Java program that reads a square matrix of integers from stdin and finds the sum of its diagonal elements. **Input Format:** - Line 1: The si...

Your Answer:

```
import java.util.Scanner;

public class DiagonalSum {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();

        int[][] matrix = new int[n][n];
        for (int i = 0; i < n; i++)
            for (int j = 0; j < n; j++)
                matrix[i][j] = scanner.nextInt();

        scanner.close();

        int sum = 0;
        for (int i = 0; i < n; i++)
            sum += matrix[i][i];  // main diagonal: top-left to bottom-right

        System.out.println(sum);
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int size = scanner.nextInt();
        int[][] matrix = new int[size][size];
        for (int i = 0; i < size; i++) {
            for (int j = 0; j < size; j++) {
                matrix[i][j] = scanner.nextInt();
            }
        }
        int sum = 0;
        for (int i = 0; i < size; i++) {
            sum += matrix[i][i];
        }
        System.out.println(sum);
        scanner.close();
    }
}
```

■ *The student's code failed to pass one out of five test cases, indicating that the logic for calculating the diagonal sum is mostly correct but may not handle a specific edge case. The issue might be related to the input or matrix size, but without more information, it's difficult to pinpoint the exact cause. The code seems to correctly calculate the sum of the main diagonal elements, so the problem may lie in how it handles certain input scenarios.*

■ Q3. Write a Java program that reads a string from stdin and checks if it is a palindrome, ignoring case and non-alphanumeric characters. **Input Format:*...

Your Answer:

```
import java.util.Scanner;

public class PalindromeChecker {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine().trim();
        scanner.close();

        // Keep only alphanumeric characters, convert to lowercase
        String cleaned = input.replaceAll("[^a-zA-Z0-9]", "").toLowerCase();

        // Check palindrome with two pointers
        int left = 0, right = cleaned.length() - 1;
        boolean isPalindrome = true;

        while (left < right) {
            if (cleaned.charAt(left) != cleaned.charAt(right)) {
                isPalindrome = false;
                break;
            }
            left++;
            right--;
        }

        System.out.println(isPalindrome ? "Palindrome" : "Not Palindrome");
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String cleanInput = input.replaceAll("[^a-zA-Z0-9]", "").toLowerCase();
        String reversedInput = new StringBuilder(cleanInput).reverse().toString();
        if (cleanInput.equals(reversedInput)) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not Palindrome");
        }
        scanner.close();
    }
}
```

■ *The student's code failed one test case due to a Runtime Error, indicating that the code encountered an exception or error during execution. This suggests that the issue lies in the logic or handling of certain inputs, rather than syntax or incomplete code. The exact cause of the error is not specified, but it may be related to the way the code handles edge cases or specific input scenarios.*

■ Q4. Write a Java program that reads a string from stdin and looks up the corresponding value in a map. **Input Format:** - Line 1: A string - Line 2: The...

Your Answer:

```
import java.util.Scanner;
import java.util.HashMap;
import java.util.Map;

public class MapLookup {

    public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);

        String key     = scanner.nextLine().trim();
        String entries = scanner.nextLine().trim();

        Map<String, String> map = new HashMap<>();

        for (String pair : entries.split("\\s+")) {
            String[] kv = pair.split(":", 2);
            if (kv.length == 2) {
                map.put(kv[0], kv[1]);
            }
        }

        scanner.close();

        String result = map.get(key);
        System.out.println(result != null ? result : "Key not found");
    }
}
```

Correct Answer:

```
import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        Map<String, String> map = new HashMap<>();
        String[] entries = scanner.nextLine().split(" ");

        for (String entry : entries) {
            String[] keyValue = entry.split(":");
            map.put(keyValue[0], keyValue[1]);
        }

        System.out.println(map.get(input));
        scanner.close();
    }
}
```

■ *The code failed to find the key 'foo' in the map, resulting in the output 'Key not found' instead of the expected 'bar'. This suggests that the key-value pairs are not being split correctly or the map is not being populated as expected. The issue likely lies in how the code handles the input string 'foo' and the map entries 'hello:world foo:bar'.*

■ Q5. Write a program that reads a list of integers and prints the filtered list of even numbers. **Input Format:** - Line 1: Number of integers (integer) ...

Your Answer:

```
import java.util.Scanner;
import java.util.ArrayList;
import java.util.List;

public class EvenFilter {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = Integer.parseInt(scanner.nextLine().trim());
        String[] tokens = scanner.nextLine().trim().split("\\s+");
```

```
        scanner.close();

        List<String> evens = new ArrayList<>();
        for (int i = 0; i < n; i++) {
            int num = Integer.parseInt(tokens[i]);
            if (num % 2 == 0) {
                evens.add(String.valueOf(num));
            }
        }

        System.out.println(String.join(" ", evens));
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        String[] input = scanner.nextLine().split(" ");
        StringBuilder evenNumbers = new StringBuilder();

        for (String num : input) {
            int number = scanner.nextInt();
            if (number % 2 == 0) {
                evenNumbers.append(number).append(" ");
            }
            scanner = new Scanner(System.in); // Reset scanner
        }

        // Corrected logic
        scanner = new Scanner(System.in);
        scanner.nextLine(); // Consume newline left-over
        String[] nums = scanner.nextLine().split(" ");
        for (String num : nums) {
            int number = Integer.valueOf(num);
            if (number % 2 == 0) {
                evenNumbers.append(number).append(" ");
            }
        }

        System.out.println(evenNumbers.toString().trim());
        scanner.close();
    }
}
```

■ *The student's code failed due to a syntax error, as indicated by the error message "SyntaxError: invalid syntax" on line 5. This suggests that the Python interpreter is being used to run Java code, which is causing the syntax error. The code appears to be written in Java, but the test environment is expecting Python code.*

■ Recommendations

Areas that need improvement:

- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.